

# Specification, Estimation & Costing

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

**Course Code**

4531

**Credits**

4

**Periods**

5/week

## Course Details

**Title:** Specification, Estimation & Costing

**Department:** Common

**Revision:** REV\_Specification, Estimation & Costing

**Pattern:** Theory / Practical

Complies with official SITTTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

## Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 4531 — Specification, Estimation & Costing**. Use the official SITTTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

**Malayalam support:** ു ു൬൬൬൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬൬൬൬ English technical terms- ു൬൬൬൬൬൬൬൬൬. ു൬ ു൬൬൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬൬൬൬, formula, diagram, units, definitions ു൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬.

**Common mistakes and precaution:** Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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#### OBJECTIVES

## Course Objectives and Outcomes

### Objectives

1. Provide deep technical understanding of Specification, Estimation & Costing.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

### CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

## MODULE 1

# Core Principles

## Outcomes

Master foundational terminology and core definitions of Specification, Estimation & Costing.

## Study

Detailed analysis of core principles and theoretical foundations.

**Malayalam Support Note:** ഘടനാപരമായ അടിസ്ഥാനപദങ്ങളും നിർവ്വചനങ്ങളും സ്പെസിഫിക്കേഷൻ, എസ്റ്റിമേഷൻ, കോസ്റ്റിംഗ് എന്നിവയുടെ അടിസ്ഥാനപദങ്ങളും പരിചയപ്പെടുക.

## MODULE 2

# Applied Methods

## Outcomes

Apply standard calculation techniques and operational procedures.

## Study

Numerical problem-solving techniques and practical workflows.

**Malayalam Support Note:** സ്റ്റാൻഡർഡ് കണക്കുകൂട്ടൽ രീതികളും പ്രവർത്തനപരമായ പ്രക്രിയകളും പ്രയോഗിക്കുക.

## MODULE 3

# Advanced Integration

## Outcomes

Evaluate complex systems and optimize performance.

## Study

Advanced system integration and troubleshooting methodologies.

**Malayalam Support Note:** ഐക്യരൂപം സാങ്കേതികവിദ്യയുടെ ഉപയോഗം വഴി സങ്കീർണ്ണ സംവിധാനങ്ങളെ മെച്ചപ്പെടുത്തുന്നതിനും പരിശോധനയ്ക്കും ഉപയോഗിക്കുന്നു.

## MODULE 4

# Synthesis & Case Studies

## Outcomes

Design comprehensive technical solutions and demonstrate mastery.

## Study

Real-world case studies and industry project design.

**Malayalam Support Note:** ഐക്യരൂപം സാങ്കേതികവിദ്യയുടെ ഉപയോഗം വഴി സങ്കീർണ്ണ സംവിധാനങ്ങളെ മെച്ചപ്പെടുത്തുന്നതിനും പരിശോധനയ്ക്കും ഉപയോഗിക്കുന്നു.

# Practice Model Question Paper

Time: 3 Hours | Max Marks: 75

## Part A (2 marks each)

1. Define core concepts of Specification, Estimation & Costing.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

## Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

# Full Answer Key

**Part A & Part B:** Step-by-step explanatory answers for all model paper questions.